Week 09 teacher guide

Data, center, and spread

PowerPoint slides 765-868 | Student workbook pp. 70-78

Data vary. Center describes a typical value, and spread describes the variation. A display makes both easier to see.

The seven days

Day 057 · p. 70 · Saturday · 20 questions,including 2 challenges
Data displays and center

Day 058 · pp. 71-73 · Sunday · 20 questions,including 2 challenges
Spread and box plots

Day 059 · p. 74 · Monday · 20 questions,including 2 challenges
Data, center, and spread

Day 060 · p. 75 · Tuesday · 20 questions,including 2 challenges
Data, center, and spread

Day 061 · p. 76 · Wednesday · 25 questions,including 2 challenges
Data, center, and spread

Day 062 · p. 77 · Thursday · 20 questions,including 2 challenges
Data, center, and spread

Day 063 · p. 78 · Friday · 20 questions,including 2 challenges
Weekly check

Use one day at a time

The student pages contain questions only. Work on separate plain paper, labeling the Day number and item IDs. Weekend lessons use 20 core questions including C1 and C2. Weekdays use 20 questions; Wednesday uses 25. Allow 30-45 minutes on ordinary weekdays and 40-55 minutes on Wednesday. Split a demanding day into two sittings if needed.

The Sunday reference page is for checking a method after an attempt. Keep answer-review slides out of view until the corresponding work is complete. Weekly answer-review sections are optional after Friday and do not add to the 80-minute weekend lesson.

Built-in review

Day 059: Data and statistics (9, 10, 11, 12)

Day 060: Data and statistics (9, 10, 11, 12)

Day 061: Weekly puzzle (13)

Day 062: Number lines and coordinates (9, 10, 11, 12)

On Friday, record routine accuracy out of 18 separately from the two challenge explanations. For each challenge, note whether the learner connected the steps, used a valid model or calculation, and justified the conclusion. Revisit a missed model, then check with a fresh item later.

Assessment alternatives: Grade 6 cumulative check (p. 72). Replace ordinary practice or use a separate assessment sitting; do not add these on top of the assigned day.

Saturday. Data displays and center

Day 057 · p. 70 | Slides 765-794

Goal: Read data displays and use mean or median to describe a typical value.

Check first: Ordering numbers, division, and reading a scale.

Materials: pencil and plain paper; use the projected models. Sketch strips, grids, or nets as needed; make any manipulatives from plain paper.

80 minutes: recall 7; launch and explore 18; model and guided practice 20; compare methods 10; challenges 20; summary and exit 5. Allow up to 10 extra minutes for discussion. Assessment alternatives replace routine practice.

Follow the numbered slide order below. The timing budgets span the named phases even when guided practice or discussion returns later. On busy weekends, split the lesson into two sittings and resume at the next slide.

Teaching sequence

Slides 765-766 · Orient

Read the goal. Ask what is familiar; name the quantity or relationship in focus.

Slides 767-768 · Recall · R1, R2, R3, R4

Try from memory, then check. Use one error for a quick reminder; keep this brief.

Slides 769-770 · Explore · P1, P2

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 771 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 772-774 · Explore · P3, P4, P5, P6, P7

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 775-777 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 778 · Guided practice · F

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 779 · Compare methods

Connect diagram and calculation. Ask where each quantity appears and why both methods preserve it.

Slides 780 · Discuss an error · D

Locate the first incorrect step. Explain why it is tempting; repair it with a model or counterexample.

Slides 781 · Language

Attach each term to a model. Ask for an example and a non-example.

Slides 782-784 · Guided practice · X1, X2, X3

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 785-786 · Deepening

Use the investigation below. Predict, calculate, and explain the structural reason. Budget about 8 minutes.

Slides 787-790 · Challenges · C1, C2

Budget about 10 minutes per challenge. Require linked reasoning; give a single hint after an independent attempt.

Slides 791 · Summary · S

Ask for one sentence or annotated example in the learner’s own words. Listen for the main relationship.

Slides 792 · Exit · E

Keep answers hidden. Collect an independent response and its reasoning before helping.

Slides 793 · Weekday bridge

Point to the next five days: 18 routines plus 2 challenges, or 23 routines plus 2 challenges on Wednesday.

Slides 794 · Tutor check

After the exit, use the hidden tutor slide to identify secure understanding, a small repair, or needed re-teaching.

Saturday. Discussion and support

Investigation T: A balanced total can hide an unusual value

Watch and discuss; this is a projected investigation, not an extra assigned question.

Compare data A: 4, 5, 6, 7, 8 and B: 4, 5, 6, 7, 28. Predict which center changes. Find both means and medians, and use a dot plot to explain why the changed mean moves by 4 although only one observation moves by 20.

1. The changed value adds 20 to the total, not to each observation.
2. Dividing that increase among 5 values raises the mean by 4.
3. The middle position still contains 6, so the median stays fixed.

Ask: How is one 20-unit change shared in a five-value mean?

If stuck: Write both totals and keep the denominator 5 visible.

Expected: A mean 6, median 6; B mean 10, median 6. The mean increases 20 ÷ 5 = 4.

C1 · Challenge discussion

Ask: Which center moves when only the far-right value moves?

If stuck: Compare the total and the middle position separately.

Expected: Original mean 12, median 10; new mean 10, median 10. Mean changes 2; median changes 0. Original median 10 better describes the cluster.

C2 · Challenge discussion

Ask: Why does adding a value equal to the mean preserve the mean?

If stuck: Find the required total before the missing value.

Expected: Missing 14; median 14; after adding 14, both mean and median remain 14.

What to notice

Uses one dot per observation, including repeats. → Check that dot count equals data count.

Orders values before finding the middle. → Ask which values are on each side.

Common error: Omits repeated values in a dot plot. Repair: Cross out each raw value after plotting one dot.

Sunday. Spread and box plots

Day 058 · pp. 71-73 | Slides 795-826

Goal: Compare how data vary using range, IQR, and mean absolute deviation.

Check first: Ordering data, mean, median, and absolute value.

Materials: pencil and plain paper; use the projected models. Sketch strips, grids, or nets as needed; make any manipulatives from plain paper.

80 minutes: recall 7; launch and explore 18; model and guided practice 20; compare methods 10; challenges 20; summary and exit 5. Allow up to 10 extra minutes for discussion. Assessment alternatives replace routine practice.

Follow the numbered slide order below. The timing budgets span the named phases even when guided practice or discussion returns later. On busy weekends, split the lesson into two sittings and resume at the next slide.

Teaching sequence

Slides 795-796 · Orient

Read the goal. Ask what is familiar; name the quantity or relationship in focus.

Slides 797-798 · Explore · P1, P2

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 799 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 800-803 · Explore · P3, P4, P5, P6, P7

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 804-806 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 807 · Guided practice · F

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 808 · Discuss an error · D

Locate the first incorrect step. Explain why it is tempting; repair it with a model or counterexample.

Slides 809-815 · Guided practice · X1, X2, X3, X4, X5, X6, X7

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 816-817 · Deepening

Use the investigation below. Predict, calculate, and explain the structural reason. Budget about 8 minutes.

Slides 818-821 · Challenges · C1, C2

Budget about 10 minutes per challenge. Require linked reasoning; give a single hint after an independent attempt.

Slides 822 · Summary · S

Ask for one sentence or annotated example in the learner’s own words. Listen for the main relationship.

Slides 823 · Exit · E

Keep answers hidden. Collect an independent response and its reasoning before helping.

Slides 824 · Weekday bridge

Point to the next five days: 18 routines plus 2 challenges, or 23 routines plus 2 challenges on Wednesday.

Slides 825 · Tutor check

After the exit, use the hidden tutor slide to identify secure understanding, a small repair, or needed re-teaching.

Slides 826 · Assessment alternative

Replace ordinary practice with this check, or use a separate sitting. It is not extra assigned work.

Sunday. Discussion and support

Investigation T: Same range, different middle half

Watch and discuss; this is a projected investigation, not an extra assigned question.

Compare A: 0, 1, 2, 4, 6, 8, 9, 10 and B: 0, 4, 4, 5, 5, 6, 6, 10. Compute each median, range, and IQR. Draw the box plots on the same scale and explain what the box widths reveal that the ranges do not.

1. Order and halve the observations: four values in each half.

2. Find the halves'middle pairs to get Q1 and Q3.

3. Compare IQR 7 with IQR 2; matching endpoints did not imply matching spread in the middle.

Ask: Which observations determine the box, and which determine the whiskers?

If stuck: Mark the two halves by count, not by cutting the value line in half.

Expected: Both median 5 and range 10; A Q1 = 1.5, Q3 = 8.5, IQR 7; B Q1 = 4, Q3 = 6, IQR 2.

C1 · Challenge discussion

Ask: How can the total stay fixed while distances grow?

If stuck: Mark the common mean 5 and measure each distance.

Expected: Both mean 5; A range 6, MAD 2; B range 10, MAD 3. B is more spread out.

C2 · Challenge discussion

Ask: Does moving all values change the gaps between them?

If stuck: Translate every landmark the same 3 units.

Expected: Original (2, 4, 8, 12, 14), IQR 8, range 12; new (5, 7, 11, 15, 17), IQR 8, range 12.

What to notice

Splits the ordered list by count. → Count the values in each half.

Uses distances for MAD. → Ask why every gap is nonnegative.

Common error: Splits the range into four equal lengths. Repair: Place each value on a card and divide the cards.

Slide-order index

Follow this numerical order. The session plans explain how to teach each phase; full scripts remain in PowerPoint speaker notes. Item IDs match the student workbook.

Saturday

765. Week 9, Saturday
Watch and discuss

766. Today
Watch and discuss

767. Recall
Day 057 · Workbook p. 70 · R1, R2, R3, R4

768. Recall: answers
Day 057 · Workbook p. 70 · R1, R2, R3, R4

769. Statistical versus non-statistical questions
Day 057 · Workbook p. 70 · P1

770. Dot plot from data
Day 057 · Workbook p. 70 · P2

771. The model
Watch and discuss

772. Histogram with specified nonoverlapping bins
Day 057 · Workbook p. 70 · P3

773. Mean as fair share and balance point
Day 057 · Workbook p. 70 · P4, P5

774. Outlier effect
Day 057 · Workbook p. 70 · P6, P7

775. Worked example · 1 of 3
Watch and discuss · worked example

776. Worked example · 2 of 3
Watch and discuss · worked example

777. Worked example · 3 of 3
Watch and discuss · worked example

778. Your turn
Day 057 · Workbook p. 70 · F

779. Two ways to see it
Watch and discuss · compare the same example

780. A claim to check
Day 057 · Workbook p. 70 · D

781. Words and meaning
Watch and discuss

782. Guided practice · X1
Day 057 · Workbook p. 70 · X1

783. Guided practice · X2
Day 057 · Workbook p. 70 · X2

784. Guided practice · X3
Day 057 · Workbook p. 70 · X3

785. A balanced total can hide an unusual value
Watch and discuss · draw on separate paper

786. A balanced total can hide an unusual value
Watch and discuss · draw on separate paper

787. Challenge C1
Day 057 · Workbook p. 70 · C1

788. Challenge C1 · Discuss
Day 057 · Workbook p. 70 · C1

789. Challenge C2
Day 057 · Workbook p. 70 · C2

790. Challenge C2 · Discuss
Day 057 · Workbook p. 70 · C2

791. In your own words
Day 057 · Workbook p. 70 · S

792. Exit ticket
Day 057 · Workbook p. 70 · E

793. The week ahead
Days 059-063 · Workbook pp. 74-78

794. Tutor reference
Tutor only;hidden in slide show

Sunday

795. Week 9, Sunday
Watch and discuss

796. Today
Watch and discuss

797. Range
Day 058 · Workbook p. 71 · P1

798. Five-number summary with odd count
Day 058 · Workbook p. 71 · P2

799. The model
Watch and discuss

800. Five-number summary with even count
Day 058 · Workbook p. 71 · P3

801. MAD computed step by step
Day 058 · Workbook p. 71 · P4

802. Read box plot
Day 058 · Workbook p. 71 · P5

803. Same center
Day 058 · Workbook p. 71 · P6, P7

804. Worked example · 1 of 3
Watch and discuss · worked example

805. Worked example · 2 of 3
Watch and discuss · worked example

806. Worked example · 3 of 3
Watch and discuss · worked example

807. Your turn
Day 058 · Workbook p. 71 · F

808. A claim to check
Day 058 · Workbook p. 71 · D

809. Guided practice · X1
Day 058 · Workbook p. 71 · X1

810. Guided practice · X2
Day 058 · Workbook p. 71 · X2

811. Guided practice · X3
Day 058 · Workbook p. 71 · X3

812. Guided practice · X4
Day 058 · Workbook p. 71 · X4

813. Guided practice · X5
Day 058 · Workbook p. 71 · X5

814. Guided practice · X6
Day 058 · Workbook p. 71 · X6

815. Guided practice · X7
Day 058 · Workbook p. 71 · X7

816. Same range, different middle half
Watch and discuss · draw on separate paper

817. Same range, different middle half
Watch and discuss · draw on separate paper

818. Challenge C1
Day 058 · Workbook p. 71 · C1

819. Challenge C1 · Discuss
Day 058 · Workbook p. 71 · C1

820. Challenge C2
Day 058 · Workbook p. 71 · C2

821. Challenge C2 · Discuss
Day 058 · Workbook p. 71 · C2

822. In your own words
Day 058 · Workbook p. 71 · S

823. Exit ticket
Day 058 · Workbook p. 71 · E

824. The week ahead
Days 059-063 · Workbook pp. 74-78

825. Tutor reference
Tutor only;hidden in slide show

826. Optional assessment · Grade 6 cumulative check
Day 058 · Workbook pp. 72-72 · Grade 6 cumulative check

Answer review

827. Week 9 answers
Next lesson: slide 869. Use this review section after Friday.

828. Data displays and center · 1 of 5
Day 057 · Workbook p. 70 · Items R1, R2, R3, R4, P1

829. Data displays and center · 2 of 5
Day 057 · Workbook p. 70 · Items P2, P3, P4, P5, P6

830. Data displays and center · 3 of 5
Day 057 · Workbook p. 70 · Items P7, F, D, X1, X2

831. Data displays and center · 4 of 5
Day 057 · Workbook p. 70 · Items X3, S, E, C1

832. Data displays and center · 5 of 5
Day 057 · Workbook p. 70 · Items C2

833. Spread and box plots · 1 of 5
Day 058 · Workbook p. 71 · Items P1, P2, P3, P4

834. Spread and box plots · 2 of 5
Day 058 · Workbook p. 71 · Items P5, P6, P7, F

835. Spread and box plots · 3 of 5
Day 058 · Workbook p. 71 · Items D, X1, X2, X3, X4

836. Spread and box plots · 4 of 5
Day 058 · Workbook p. 71 · Items X5, X6, X7, S

837. Spread and box plots · 5 of 5
Day 058 · Workbook p. 71 · Items E, C1, C2

838. Grade 6 cumulative check · 1 of 5
Day 058 · Workbook p. 72 · Items 1, 2, 3, 4, 5

839. Grade 6 cumulative check · 2 of 5
Day 058 · Workbook p. 72 · Items 6, 7, 8, 9

840. Grade 6 cumulative check · 3 of 5
Day 058 · Workbook p. 72 · Items 10, 11, 12, 13, 14

841. Grade 6 cumulative check · 4 of 5
Day 058 · Workbook p. 72 · Items 15, 16, 17, 18, 19

842. Grade 6 cumulative check · 5 of 5
Day 058 · Workbook p. 72 · Items 20, 21, 22, 23, 24

843. Data, center, and spread · 1 of 5
Day 059 · Workbook p. 74 · Items 1, 2, 3, 4, 5

844. Data, center, and spread · 2 of 5
Day 059 · Workbook p. 74 · Items 6, 7, 8, 9, 10

845. Data, center, and spread · 3 of 5
Day 059 · Workbook p. 74 · Items 11, 12, 13, 14, 15

846. Data, center, and spread · 4 of 5
Day 059 · Workbook p. 74 · Items 16, 17, 18, C1

847. Data, center, and spread · 5 of 5
Day 059 · Workbook p. 74 · Items C2

848. Data, center, and spread · 1 of 5
Day 060 · Workbook p. 75 · Items 1, 2, 3, 4, 5

849. Data, center, and spread · 2 of 5
Day 060 · Workbook p. 75 · Items 6, 7, 8, 9, 10

850. Data, center, and spread · 3 of 5
Day 060 · Workbook p. 75 · Items 11, 12, 13, 14, 15

851. Data, center, and spread · 4 of 5
Day 060 · Workbook p. 75 · Items 16, 17, 18, C1

852. Data, center, and spread · 5 of 5
Day 060 · Workbook p. 75 · Items C2

853. Data, center, and spread · 1 of 6
Day 061 · Workbook p. 76 · Items 1, 2, 3, 4, 5

854. Data, center, and spread · 2 of 6
Day 061 · Workbook p. 76 · Items 6, 7, 8, 9, 10

855. Data, center, and spread · 3 of 6
Day 061 · Workbook p. 76 · Items 11, 12, 13, 14, 15

856. Data, center, and spread · 4 of 6
Day 061 · Workbook p. 76 · Items 16, 17, 18, 19, 20

857. Data, center, and spread · 5 of 6
Day 061 · Workbook p. 76 · Items 21, 22, 23, C1

858. Data, center, and spread · 6 of 6
Day 061 · Workbook p. 76 · Items C2

859. Data, center, and spread · 1 of 5
Day 062 · Workbook p. 77 · Items 1, 2, 3, 4, 5

860. Data, center, and spread · 2 of 5
Day 062 · Workbook p. 77 · Items 6, 7, 8, 9, 10

861. Data, center, and spread · 3 of 5
Day 062 · Workbook p. 77 · Items 11, 12, 13, 14, 15

862. Data, center, and spread · 4 of 5
Day 062 · Workbook p. 77 · Items 16, 17, 18, C1

863. Data, center, and spread · 5 of 5
Day 062 · Workbook p. 77 · Items C2

864. Weekly check · 1 of 5
Day 063 · Workbook p. 78 · Items 1, 2, 3, 4, 5

865. Weekly check · 2 of 5
Day 063 · Workbook p. 78 · Items 6, 7, 8, 9, 10

866. Weekly check · 3 of 5
Day 063 · Workbook p. 78 · Items 11, 12, 13, 14, 15

867. Weekly check · 4 of 5
Day 063 · Workbook p. 78 · Items 16, 17, 18, C1

868. Weekly check · 5 of 5
Day 063 · Workbook p. 78 · Items C2

Complete weekly answers

Match Day,item ID,and page before checking. Equivalent correct forms and valid alternative methods are welcome. Computational keys are compact; challenges and conceptual items include reasoning. No challenge is complete with an unsupported answer alone.

Day 057 · Data displays and center

Workbook p. 70 · assigned set

R1 · E-RELATE

(1, 2), (3, 6), (5, 10)

R2 · G-SA

88 square cm

R3 · N-DIST

5 units

R4 · E-ADD

x = 7

P1 · S-QUESTION

The shelf question; page counts can vary across books.

Reason: The first asks for one fixed book. The second expects several different values.

P2 · S-DISPLAY

Seven dots; counts 1, 2, 1, 2, 1 at 1 through 5. Symmetric around 3.

Reason: Stack one dot for each student at that student's number of books.

P3 · S-DISPLAY

Bin counts: 3, 3, 2

P4 · S-MEAN

Mean 5; deviations −3, −1, 0, 4 sum to 0

Reason: There are 20 counters for four jars. Move four from the jar of 9 to fill the two low jars.

P5 · S-MEDIAN

(a) 4; (b) 5

P6 · S-CENTER

Mean: 4 to 8. Median: 4 to 4. The median changes less.

Reason: The outlier increases the total by 20, so the five-value mean increases by 4. The middle value stays 4.

P7 · S-MEAN

9

F · S-MEDIAN

6; 4.5

Reason: Check each step against the projected worked example; require the learner to explain the operation or relationship.

D · S-MEDIAN

Order 2, 3, 6, 9, then average 3 and 6. Median = 4.5.

Reason: Finds the displayed middle before ordering. Put the values on separate cards and order them.

X1 · S-QUESTION

B; it anticipates varying page counts across books.

Reason: A asks for one fixed count; B gathers a distribution of values that may differ.

X2 · S-DISPLAY

7 dots; most frequent 3 with 3 dots.

Reason: Place one dot per observation; stack repeated values at the same location.

X3 · S-DISPLAY

2, 3, 3.

S · EXPLAIN

Answers vary. A statistical question expects different answers. Mean shares the total equally; median locates the ordered middle.

Reason: Accept an accurate explanation supported by a matching example.

E · S-MEDIAN

Ordered: 2, 4, 5, 6, 8, 11. Median 5.5; mean 6.

Reason: The middle values are 5 and 6. Their mean is 5.5. The total is 36, and 36 ÷ 6 = 6.

C1 · S-CENTER

Original mean 12, median 10; new mean 10, median 10. Mean changes 2; median changes 0. Original median 10 better describes the cluster.

Reason: Original sum 84 gives mean 12. The ordered fourth value is 10. Replacing 24 by 10 lowers the sum by 14 to 70, so the mean becomes 10. The median stays 10. The unusually high 24 pulls the mean above most observations.

C2 · S-MEAN

Missing 14; median 14; after adding 14, both mean and median remain 14.

Reason: The five-score total is 5 × 14 = 70. Known scores total 56, leaving 14. Ordered data 10, 12, 14, 16, 18 have middle 14. Add 14: total 84 over 6 gives 14, and middle pair 14, 14 also gives 14.

Day 058 · Spread and box plots

Workbook p. 71 · assigned set

P1 · S-RANGE

9

P2 · S-IQR

Minimum 1; Q1 3; median 6; Q3 9; maximum 11

P3 · S-IQR

1, 3, 7, 11, 13; IQR 8

P4 · S-MAD

Mean 5; distances 3, 1, 1, 3; MAD 2

P5 · S-BOX

Upper quarter: 9 to 12. No; each quarter represents about 25% of the data.

Reason: The plot groups data by ordered positions, not equal numerical lengths.

P6 · S-SPREAD

Both means 5. MAD A = 1, MAD B = 3. B is more spread out.

Reason: B's values lie farther from the shared mean on average.

P7 · S-SPREAD

Range 10 to 30; IQR stays 4.5

Reason: Q1 is 3.5 and Q3 is 8 in both sets. The maximum changes, but these middle-half markers do not.

F · S-MAD

1; 1; 2

Reason: Check each step against the projected worked example; require the learner to explain the operation or relationship.

D · S-MAD

The distances 3, 1, 1, 3. Their mean is 2.

Reason: Signed deviations cancel instead of measuring distance. Measure each gap from the mean with a ruler.

X1 · S-RANGE

10; the minimum and maximum.

Reason: $12 - 2 = 10$. Other values do not affect this calculation.

X2 · S-IQR

(2, 3, 7, 11, 13); IQR 8.

X3 · S-IQR

$(2, \frac{9}{2}, 8, \frac{23}{2}, 14)$; IQR 7.

X4 · S-MAD

Mean 6; distances 4, 2, 2, 4; MAD 3.

Reason: The distances total 12; divide by 4 to get 3. Use absolute distances, not signed changes.

X5 · S-BOX

Range 12; IQR 7; upper quarter from 10 to 13.

Reason: Range = $13 - 1$; IQR = $10 - 3$. The upper quarter is between Q3 and the maximum.

X6 · S-MAD

A: MAD 1; B: MAD 3; B more spread out.

Reason: Average the four distances from the common mean. Greater average distance means greater dispersion.

X7 · S-SPREAD

Range 12→32; IQR 8→8.

Reason: Only the maximum changes. The lower half median 4 and upper half median 12 stay fixed.

S · EXPLAIN

Answers vary. Center describes a typical value. Spread describes how far values are apart or away from the center.

Reason: Accept an accurate explanation supported by a matching example.

E · S-IQR

Q1 3; median 6; Q3 10; IQR 7

Reason: Lower half: 1, 3, 3, 5. Upper half: 7, 9, 11, 13. Use each half's median.

C1 · S-MAD

Both mean 5; A range 6, MAD 2; B range 10, MAD 3. B is more spread out.

Reason: A distances from 5 are 3, 1, 1, 3, total 8, MAD 2. B distances are 5, 1, 1, 5, total 12, MAD 3. The extreme values moved equally away from 5, keeping the mean but increasing spread.

C2 · S-IQR

Original (2, 4, 8, 12, 14), IQR 8, range 12; new (5, 7, 11, 15, 17), IQR 8, range 12.

Reason: Original median 8; lower half 2, 4, 6 gives Q1 = 4, and upper half 10, 12, 14 gives Q3 = 12. Adding 3 shifts every landmark equally. Differences $12 - 4$ and $14 - 2$ are unchanged after the shift.

Day 058 · Grade 6 cumulative check

Workbook p. 72 · assessment alternative

1 · R-RATIO

5 : 2; 2 : 5

Reason: Keep the requested first color first.

2 · R-TABLE
21

3 · R-RATE
24 miles

4 · P-PART
42 points

5 · P-WHOLE
40 shells

6 · F-DIV
$1\frac{1}{6}$ groups

7 · F-DIV
$\frac{5}{24}$ liter

8 · F-DIV
$\frac{2}{5}$

Reason: Two and one half is five halves. Five halves times two fifths equals one.

9 · D-MULT
$7.35

10 · D-DIV
1.2

11 · N-DIV
42

12 · E-DIST
2x + 14

13 · E-ADD
x = 14

14 · E-MULT
x = 16

15 · G-PARA
36 square cm

16 · G-TRI
18 square cm

17 · G-SA
60 square cm

18 · N-ORDER
−7, −1, 2

19 · N-REFLECT
(3, 4)

20 · E-INEQ
$x \geq 9$; yes

21 · S-MEAN
5

22 · S-MEDIAN
5

23 · S-IQR
Q1 3.0; Q3 9.0; IQR 6

24 · S-MAD
2

Day 059 · Data, center, and spread

Workbook p. 74 · assigned set

1 · N-FACT
33

2 · N-FACT
77

3 · N-FACT
44

4 · N-FACT
99

5 · N-REFLECT
(3, 4)

6 · R-EQUIV
No

7 · N-FACT
10

8 · S-MEAN
4

9 · S-MEAN · review
5

10 · S-MEDIAN · review
5

11 · S-MEDIAN · review
5

12 · S-MEAN · review
7

13 · S-MEDIAN
Order 2, 3, 6, 9, then average 3 and 6. Median = 4.5.

14 · S-QUESTION

B; it anticipates varying page counts across books.

Reason: A asks for one fixed count; B gathers a distribution of values that may differ.

15 · S-DISPLAY

7 dots; most frequent 5 with 3 dots.

Reason: Place one dot per observation; stack repeated values at the same location.

16 · S-DISPLAY

2, 3, 3.

17 · S-MEAN

7

Reason: The total 28 divided among 4 equal shares gives 7.

18 · S-MEDIAN

7

C1 · S-MEAN

Combined mean 10.5, not 10.

Reason: A contributes $3 \times 8 = 24$ points; B contributes $5 \times 12 = 60$. The combined 84 points belong to 8 scores, so $84 \div 8 = 10.5$. Team B has more scores and therefore contributes more to the combined mean.

C2 · S-CENTER

One answer 1, 4, 6, 9; mean 5, median 5, range 8. Second 4, 4, 6, 6 has range 2.

Reason: The first total is 20; the middle pair 4, 6 averages 5, and $9 - 1 = 8$. The second also totals 20 and has middle pair 4, 6, but $6 - 4 = 2$. Other valid constructions are acceptable.

Day 060 · Data, center, and spread

Workbook p. 75 · assigned set

1 · F-EQUIV

0.5; 50%

2 · F-EQUIV

0.25; 25%

3 · F-EQUIV

0.75; 75%

4 · F-EQUIV

0.125; 12.5%

5 · N-ABS

12 units from zero

Reason: Count the unit steps between the number and zero.

6 · D-DIV

30

7 · N-FACT

12

8 · S-MAD

3

9 · S-RANGE · review

9

10 · S-IQR · review

Q1 3.0; Q3 9.0; IQR 6

11 · S-MAD · review

2

12 · S-SPREAD · review

Both means 4. Range A 2; range B 6. B is more spread out.

Reason: Find the center, then compare the endpoint gaps.

13 · S-MAD

The distances 3, 1, 1, 3. Their mean is 2.

14 · S-QUESTION

B; it anticipates varying page counts across books.

Reason: A asks for one fixed count; B gathers a distribution of values that may differ.

15 · S-DISPLAY

7 dots; most frequent 6 with 3 dots.

Reason: Place one dot per observation; stack repeated values at the same location.

16 · S-DISPLAY

2, 3, 3.

17 · S-MEAN

8

Reason: The total 32 divided among 4 equal shares gives 8.

18 · S-MEDIAN

8

C1 · S-SPREAD

Summary (1, 3, 6, 9, 11), IQR 6; range 10→30; IQR stays 6.

Reason: Lower half 1, 2, 4, 5 gives Q1 = 3; upper half 7, 8, 10, 11 gives Q3 = 9. Replacing only the maximum by 31 leaves both half-middle pairs unchanged. The endpoints'gap grows by 20; the middle-half gap stays 6.

C2 · S-BOX

Same median 5 and range 10; IQR A6, B2; sample sizes cannot be determined.

Reason: Both plots start 0, end 10, and have center 5. The boxes span $8 - 2 = 6$ and $6 - 4 = 2$, so A's middle half is more dispersed. A five-number summary does not give the total number of data values.

Day 061 · Data, center, and spread

Workbook p. 76 · assigned set

1 · N-SCALE
112

2 · N-SCALE
11

3 · N-SCALE
23

4 · N-SCALE
33

5 · S-RANGE
10

6 · S-MAD
$\frac{3}{2}$

7 · N-ABS
3 units from zero

Reason: Count the unit steps between the number and zero.

8 · G-SA
84 square cm

9 · R-TABLE
21

10 · F-DIV
6 pieces

11 · N-FACT
28

12 · G-VOL
84 cm³

13 · S-MEAN · review
2, 3, 5, 8, 12 and 2, 4, 5, 7, 12

14 · S-QUESTION
B; it anticipates varying page counts across books.

Reason: A asks for one fixed count; B gathers a distribution of values that may differ.

15 · S-DISPLAY
7 dots; most frequent 7 with 3 dots.

Reason: Place one dot per observation; stack repeated values at the same location.

16 · S-DISPLAY
2, 3, 3.

17 · S-MEAN
9

Reason: The total 36 divided among 4 equal shares gives 9.

18 · S-MEDIAN
9

19 · S-MEAN
13

20 · S-RANGE
10; the minimum and maximum.

Reason: 15 − 5 = 10. Other values do not affect this calculation.

21 · S-IQR
(5, 6, 10, 14, 16); IQR 8.

22 · S-IQR
(5, $\frac{15}{2}$, 11, $\frac{29}{2}$, 17); IQR 7.

23 · S-MAD
Mean 9; distances 4, 2, 2, 4; MAD 3.

Reason: The distances total 12; divide by 4 to get 3. Use absolute distances, not signed changes.

C1 · S-DISPLAY
Frequencies 2, 3, 3; highest bin $\frac{3}{8}$; 5 belongs in 5 to under 10.

Reason: The bins contain (0, 4), (5, 5, 9), (10, 12, 14). There are 8 observations, 3 in the highest bin. Each left boundary is included and each right boundary excluded, so no observation is counted twice.

C2 · S-CENTER
Original mean 6, median 6; incorrect mean 11, median 6. The mean's jump of 5 is a warning.

Reason: Original total 36 gives 6. The error adds 30 to the total, so the mean increases 30 ÷ 6 = 5 to 11. The middle pair remains 6, 6. The mismatch between a typical middle score and the new mean suggests checking extreme entries.

Day 062 · Data, center, and spread

Workbook p. 77 · assigned set

1 · N-EST
1100

2 · N-EST
12

3 · N-EST
About 100

4 · N-EST

About 20

5 · S-RANGE

10

6 · S-MAD

$\frac{3}{2}$

7 · S-DISPLAY

2

Reason: Count every occurrence of 4.

8 · G-SA

60 square cm

9 · N-ORDER · review

−6, −3, 1

10 · N-ABS · review

5 units from zero

Reason: Count the unit steps between the number and zero.

11 · N-REFLECT · review

(2, 5)

12 · N-DIST · review

5 units

13 · S-IQR

Q1 2.5; Q3 9.5; IQR 7

14 · S-MAD

The distances 3, 1, 1, 3. Their mean is 2.

15 · R-PERCENT

$30

16 · E-EQ

x = 4

17 · S-QUESTION

B; it anticipates varying page counts across books.

Reason: A asks for one fixed count; B gathers a distribution of values that may differ.

18 · S-DISPLAY

7 dots; most frequent 8 with 3 dots.

Reason: Place one dot per observation; stack repeated values at the same location.

C1 · S-MAD

Mean 7; distances 4, 2, 0, 2, 4; MAD $\frac{12}{5}$ = 2.4.

Reason: The signed changes cancel because the mean balances the data. Spread uses nonnegative distances, so average 4 + 2 + 0 + 2 + 4 = 12 over 5 observations. Zero would incorrectly claim that all values equal the mean.

C2 · S-SPREAD

Original mean $\frac{14}{3}$, median 5, range 6; new mean $\frac{20}{3}$, median 7, range 6.

Reason: Adding 2 to each of 6 values adds 12 to the total, so the mean rises by 2. Both middle values rise 2, so the median rises 2. The maximum and minimum both rise 2, leaving their difference 6 unchanged.

Day 063 · Weekly check

Workbook p. 78 · assigned set

1 · S-RANGE

10

2 · S-MAD

$\frac{3}{2}$

3 · S-DISPLAY

2

Reason: Count every occurrence of 4.

4 · S-MEAN

10

5 · G-TRI

18 square cm

6 · N-DIST

7 units

7 · E-INEQ

$x \leq 12$; yes

8 · R-TABLE

21

9 · F-DIV

6 pieces

10 · S-MEDIAN

Order 2, 3, 6, 9, then average 3 and 6. Median = 4.5.

11 · S-QUESTION

B; it anticipates varying page counts across books.

Reason: A asks for one fixed count; B gathers a distribution of values that may differ.

12 · S-DISPLAY

7 dots; most frequent 9 with 3 dots.

Reason: Place one dot per observation; stack repeated values at the same location.

13 · S-DISPLAY

2, 3, 3.

14 · S-MEAN

11

Reason: The total 44 divided among 4 equal shares gives 11.

15 · S-MEDIAN

11

16 · S-MEAN

15

17 · S-RANGE

10; the minimum and maximum.

Reason: 17 − 7 = 10. Other values do not affect this calculation.

18 · S-IQR

(7, 8, 12, 16, 18); IQR 8.

C1 · S-MAD

Both sets mean 5, median 5, MAD 2; none changes.

Reason: The original total 20 over 4 gives 5, and middle pair 4, 6 gives 5. Distances 3, 1, 1, 3 average 2. Duplicating doubles both the total and count, and also both the total distance and count; each ratio is unchanged. The new middle pair remains 4, 6.

C2 · S-IQR

One answer 2, 4, 6, 8, 10; range 8; summary (2, 3, 6, 9, 10). Not unique: 2, 5, 6, 7, 10 also works.

Reason: The required total is 30. With 2, 6, 10 fixed, the remaining two values must total 12. Choosing 4 and 8 works. Halves 2, 4 and 8, 10 give quartiles 3 and 9. Choosing 5 and 7 preserves mean and median but changes quartiles.